TRƯỜNG ĐẠI HỌC CÔNG NGHỆ KỸ THUẬT TP. HCM
Ho Chi Minh City University of Technology and Engineering

HCM-UTE

KHOA GIAO THÔNG VÀ NĂNG LƯỢNG
Faculty of Energy Engineering and Transport

BALANCE

ON

OFF

LEFT

POWER

RIGHT

LOAD

ON

OFF

UART

ON

OFF

LCD MODE

SETTING

MONITOR

CURRENT DISCHARGE MAX

VOLTAGE DISCHARGE MIN

CURRENT CHARGE MAX

VOLTAGE CHARGE MAX

DESIGN AND DEVELOPMENT OF A BATTERY MONITORING AND WARNING SYSTEM ON AN EMBEDDED SYSTEM PLATFORM USING CAN BUS

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SYSTEM SPECIFICATIONS

MCU: STM32F103C8T6 (Cortex-M3)
Network: 1 Master + 2 Slaves, CAN 2.0A @125kbps
Battery: Li-ion 18650, 10S2P (2x5S2P); 30 - 42V, nom. 36V
Monitoring: Cell V, Total V, Vmin/Vmax/Delta, Current, Temp, SOC
Balancing: Passive
Protection: OV / UV / OC / OT
Display: 20x4 LCD (I2C)
Alert: LED + Buzzer
Telemetry: CAN + UART